

JetBrains survey of 23,262 developers worldwide

State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

A data-driven view of how language usage is shifting, what developers actually prefer, and where leaders should focus technology strategy and hiring priorities.

61%

JavaScript remains the most used language overall

35%

Python is the most common primary language

199

TypeScript leads developer satisfaction

Narrative arc

Usage shifts → Developer preference → Strategic action

Python Closes the Gap: 8 Years of Language Usage Shifts

Comparison table from the source data; 2024 usage ranking shows a reshuffled top language landscape.

Language	2017	2020	2022	2024	Trend
JavaScript	65%	70%	65%	61%	Declining from peak
Python	32%	55%	53%	57%	Steady growth
HTML/CSS	60%	61%	54%	51%	Gradual decline
SQL	42%	56%	49%	48%	Plateaued
Java	47%	54%	48%	46%	Slow decline
TypeScript	12%	28%	34%	37%	Strong growth
Go	8%	19%	19%	18%	Plateaued after growth
Rust	--	7%	9%	11%	Steady growth
PHP	30%	27%	20%	17%	Significant decline
Kotlin	2%	17%	16%	14%	Plateaued

Usage signals

70% → 61%

JavaScript fell from its 2020 peak but remains #1 overall.

32% → 57%

Python nearly closed the usage gap since 2017.

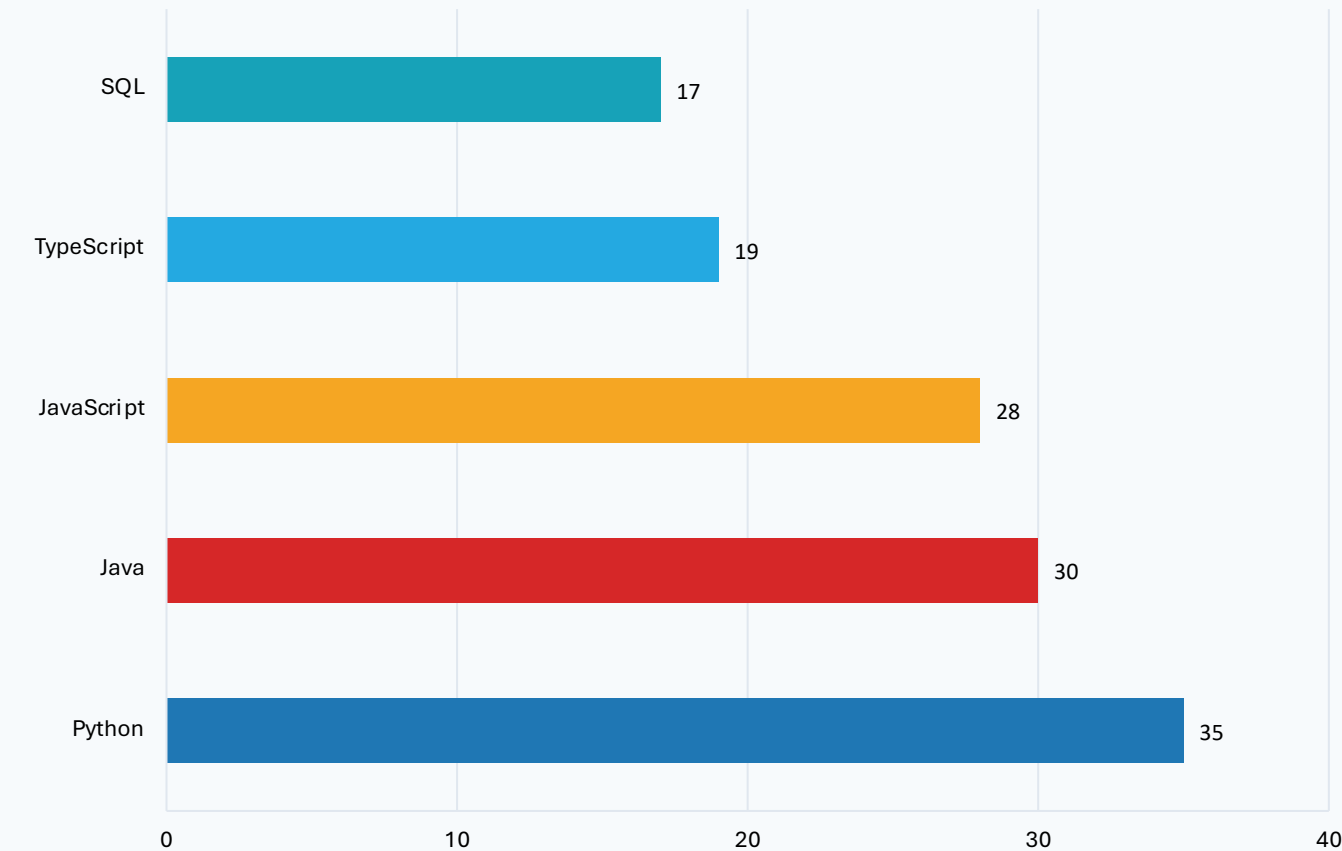
12% → 37%

TypeScript tripled its share over the period shown.

30% → 17%

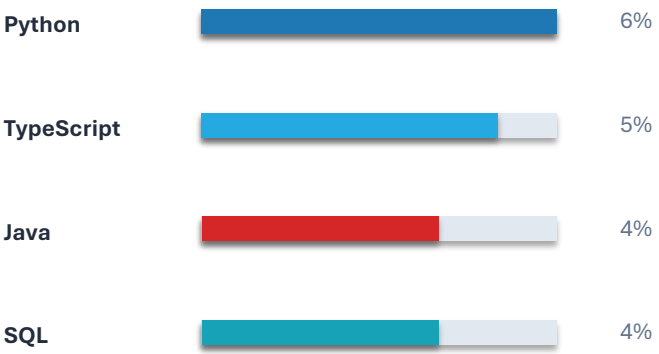
PHP shows significant long-term decline.

Python Is Now the #1 Primary Language at 35%



Lock-in is visible in adoption intent

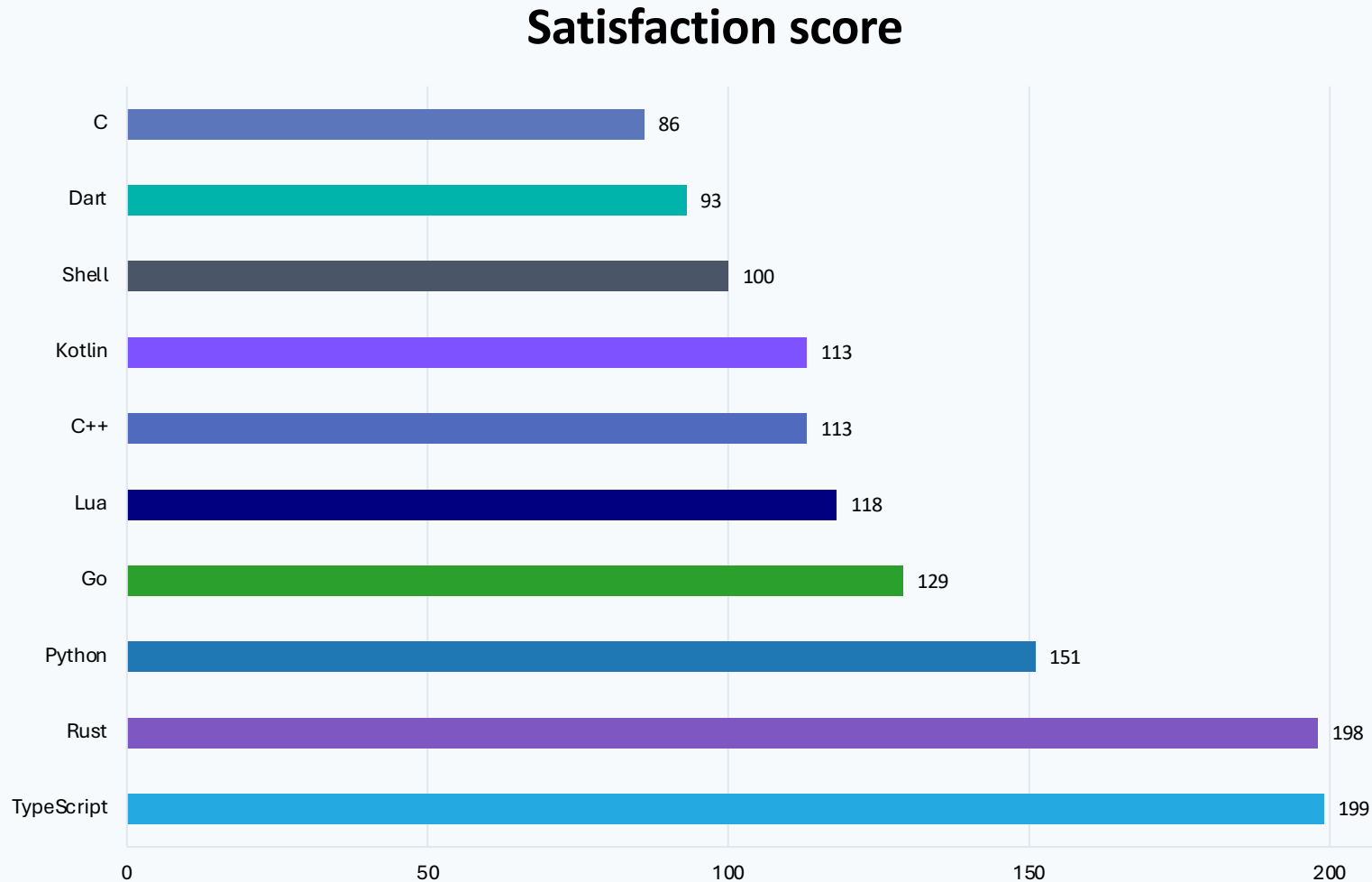
Developers show strong loyalty to their chosen ecosystems; planned adoption stays modest across languages.



Strategic implication

Talent strategy should account for ecosystem depth and switching costs, not just growth headlines.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



What the leaders have in common

199

TypeScript

Typed JavaScript ecosystem plus modern tooling.

198

Rust

Safety guarantees for performance-sensitive work.

151

Python

Broad appeal reinforced by AI and data science.

Retention lens

Satisfaction concentrates around safety, expressive type systems, and strong tooling.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Structured domain-leader table from the source brief: language choice is domain-shaped, not universally interchangeable.

Application domain	Leading language	Share	Strategic reading
AI / Machine Learning	Python	31%	Modeling, AI/ML workflows
Data Science / Analytics	Python	28%	Analysis and notebooks
Data Processing & Analytics	Python	17%	Data pipelines
User Interface	TypeScript	62%	Typed frontend scale
User Interface	JavaScript	58%	Web runtime ubiquity
APIs & Services	Java	49%	Enterprise services
Application Logic & Workflows	Java	55%	Business systems
System Tools & Components	C++	27%	Performance-critical systems
Hardware Interfacing	C++	21%	Low-level integration
Graphics Rendering / Processing	C++	15%	Graphics pipelines
Integrating with APIs & Services	Python	38%	Automation and glue code

AI + data

Python leads AI/ML, data science, data processing, and API integration.

UI

TypeScript and JavaScript together dominate user interface development.

Core systems

Java owns services and workflows; C++ owns lower-level performance domains.

Source: JetBrains State of Developer Ecosystem Report 2024; n=23,262 developers

What This Means for Your Technology Strategy in 2025

1

Invest in Python for AI/data

Python leads primary language usage at 35% and leads AI/ML, data science, data processing, and API integration domains.

2

Use TypeScript as the frontend scale bet

TypeScript leads UI share at 62% and developer satisfaction at 199.

3

Watch Rust and Go for infrastructure

Go stabilized around 18% usage; Rust continues climbing to 11% and both address modern infrastructure needs.

4

Plan legacy web-stack transitions deliberately

PHP declined from 30% in 2017 to 17% in 2024; Ruby fell from 10% to 4%.

5

Treat tooling as a retention lever

The most loved languages reflect safety, modern tooling, and expressive type systems.

Leadership takeaway Language strategy is now a portfolio decision: align domains, developer satisfaction, and hiring focus rather than standardizing on one universal stack.